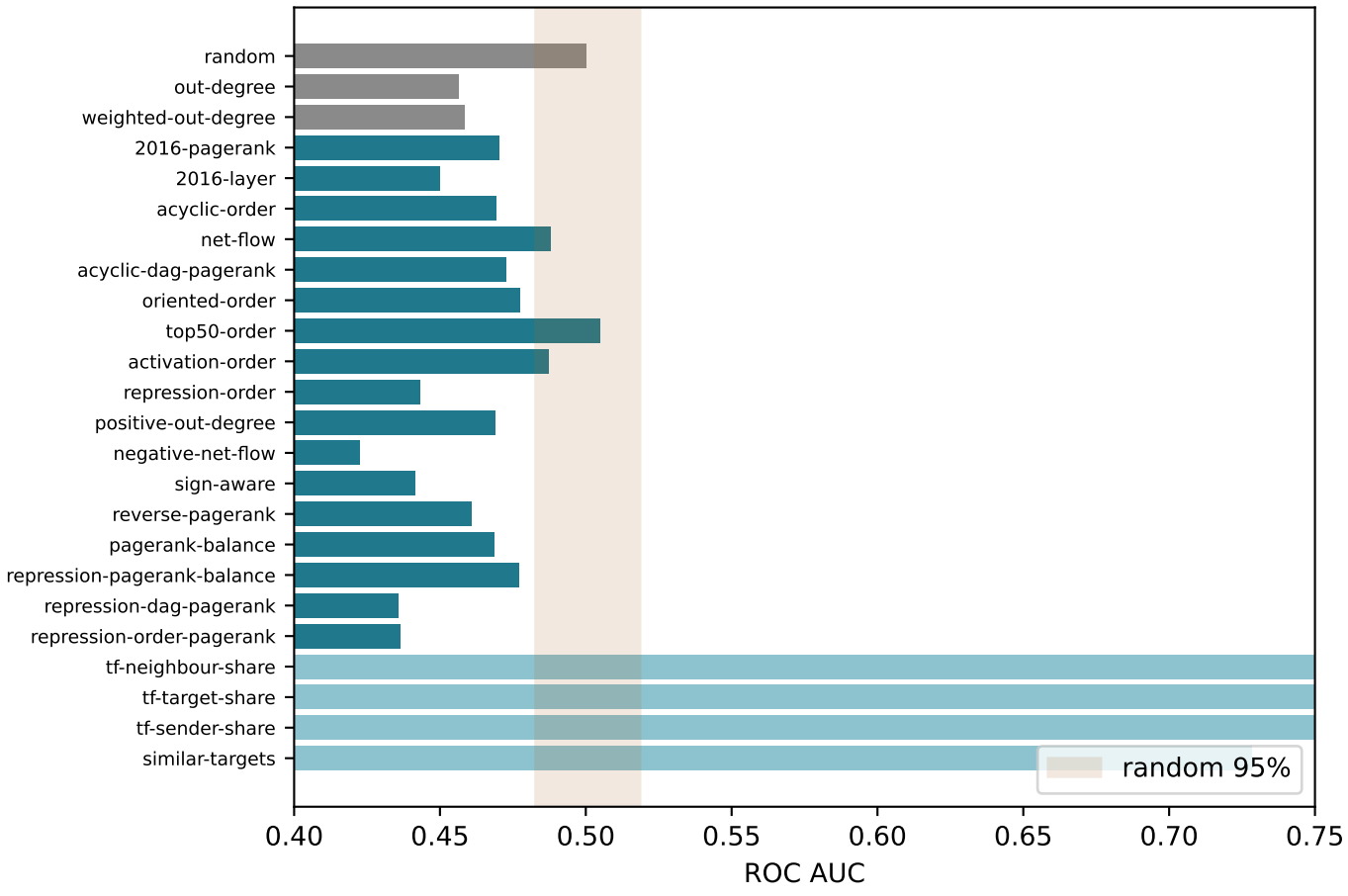
Regulator ranking against reference 'lambert-2018'

2026-11-10 · 11,992 genes ranked, 1,155 in lambert-2018-tfs.txt · top k = 100

Pipeline	Reference genes in top 100	AUC	p top-k	p AUC	AUC vs best baseline (95%)	p vs baseline
random	9.7 (4–16)	0.500 (0.483–0.519)				
out-degree	4	0.456	0.994	1.000	-0.002 (-0.003, -0.001)	< 0.001
weighted-out-degree	5	0.458	0.972	1.000	+0.000 (+0.000, +0.000)	
2016-pagerank	9	0.470	0.641	1.000	+0.012 (-0.004, +0.027)	0.142
2016-layer	8	0.450	0.765	1.000	-0.009 (-0.026, +0.009)	0.320
acyclic-order	5	0.469	0.972	1.000	+0.011 (+0.004, +0.018)	0.002
net-flow	5	0.488	0.972	0.919	+0.029 (+0.017, +0.040)	< 0.001
acyclic-dag-pagerank	2	0.473	1.000	0.999	+0.014 (+0.008, +0.020)	< 0.001
oriented-order	12	0.478	0.260	0.998	+0.019 (+0.004, +0.034)	0.010
top50-order	10	0.505	0.513	0.296	+0.046 (+0.028, +0.065)	< 0.001
activation-order	14	0.487	0.098	0.928	+0.029 (+0.020, +0.037)	< 0.001
repression-order	3	0.443	0.997	1.000	-0.015 (-0.028, -0.001)	0.026
positive-out-degree †	13	0.469	0.164	1.000	+0.010 (+0.007, +0.014)	< 0.001
negative-net-flow †	2	0.423	1.000	1.000	-0.036 (-0.050, -0.022)	< 0.001
sign-aware †	3	0.441	0.997	1.000	-0.017 (-0.024, -0.010)	< 0.001
reverse-pagerank †	6	0.461	0.934	1.000	+0.002 (-0.001, +0.006)	0.266
pagerank-balance †	13	0.469	0.164	1.000	+0.010 (-0.004, +0.024)	0.156
repression-pagerank-balance †	7	0.477	0.870	0.999	+0.019 (+0.004, +0.033)	0.016
repression-dag-pagerank †	4	0.436	0.994	1.000	-0.023 (-0.036, -0.010)	< 0.001
repression-order-pagerank †	4	0.436	0.994	1.000	-0.022 (-0.034, -0.011)	0.002
tf-neighbour-share ‡	100	0.771	< 0.001	< 0.001	+0.312 (+0.288, +0.338)	< 0.001
tf-target-share ‡	96	0.763	< 0.001	< 0.001	+0.305 (+0.281, +0.331)	< 0.001
tf-sender-share ‡	100	0.754	< 0.001	< 0.001	+0.295 (+0.271, +0.321)	< 0.001
similar-targets ‡	99	0.729	< 0.001	< 0.001	+0.270 (+0.245, +0.296)	< 0.001

Random: mean and 95% range over random orderings. p: share of random orderings scoring at least as well. Grey bars: random and baselines. AUC vs best baseline: paired bootstrap over genes (1,000 resamples). †: designed after seeing these results; needs confirming on other data. ‡: uses known TFs: scored from other genes' labels in this reference (lighter bars).



Edge checks against curated TF→target pairs

Recall ×: share of curated pairs present as edges, over chance.
Positive/negative ×: share of edges of that sign that are curated pairs, over chance. Sign: agreement with the curated sign (chance).

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Graph	Pairs	Recall (×)	Positive ×	Negative ×	Sign (chance)
raw graph	31,889	0.041 (2.71×)	3.24×	0.74×	0.84 (0.83)
resolve_reciprocals → drop_sinks → prune_triangles	31,686	0.003 (1.07×)	2.90×	0.79×	0.39 (0.38)
resolve_reciprocals → drop_sinks → prune_triangles → greedy_layering	21,591	0.002 (0.95×)	2.52×	0.85×	0.28 (0.30)
absolute	31,889	0.041 (2.71×)			
absolute → minimum_feedback_order	31,711	0.027 (2.40×)			
orient_reciprocals ratio=2.0 → absolute	31,731	0.015 (1.78×)			
orient_reciprocals ratio=2.0 → absolute → minimum_feedback_order	31,731	0.015 (1.78×)			
top_outgoing k=50 → absolute	31,889	0.016 (4.23×)			
top_outgoing k=50 → absolute → minimum_feedback_order	31,728	0.010 (3.61×)			
keep_sign sign=1 → absolute	31,889	0.038 (3.24×)			
keep_sign sign=1 → absolute → minimum_feedback_order	30,532	0.026 (2.95×)			
keep_sign sign=-1 → absolute	30,856	0.002 (0.73×)			
keep_sign sign=-1 → absolute → minimum_feedback_order	30,527	0.002 (0.80×)			
keep_sign sign=-1	30,856	0.002 (0.73×)		0.73×	0.19 (0.19)

Pipeline steps

out-degree: Most outgoing edges (raw graph)

score out_degree, descending

Graph: 2,325,173 edges (537,315 negative), largest strong component 11,992.

weighted-out-degree: Largest total outgoing weight (raw graph)

score weighted_out_degree, descending

Graph: 2,325,173 edges (537,315 negative), largest strong component 11,992.

2016-pagerank: 2016 cleaning, lowest PageRank

resolve_reciprocals() → drop_sinks() → prune_triangles() → score pagerank, ascending

Graph: 531,752 edges (474,793 negative), largest strong component 11,716.

2016-layer: 2016 cleaning, earliest greedy layer

resolve_reciprocals() → drop_sinks() → prune_triangles() → greedy_layering() → score level, ascending

Graph: 531,752 edges (474,793 negative), largest strong component 11,716; DAG keeps 311,249 edges.

acyclic-order: Earliest in acyclic order

absolute() → minimum_feedback_order() → score level, ascending

Graph: 2,325,173 edges (0 negative), largest strong component 11,992; DAG keeps 1,754,170 edges.

net-flow: Outgoing minus incoming weight

absolute() → score net_flow, descending

Graph: 2,325,173 edges (0 negative), largest strong component 11,992.

acyclic-dag-pagerank: Reverse PageRank on the acyclic DAG

absolute() → minimum_feedback_order() → score dag_reverse_pagerank, descending

Graph: 2,325,173 edges (0 negative), largest strong component 11,992; DAG keeps 1,754,170 edges.

oriented-order: Pairs with a 2x stronger side only, earliest in acyclic order

orient_reciprocals(ratio=2.0) → absolute() → minimum_feedback_order() → score level, ascending

Graph: 1,321,858 edges (0 negative), largest strong component 9,931; DAG keeps 1,309,420 edges.

top50-order: Top 50 outgoing edges per gene, earliest in acyclic order

top_outgoing(k=50) → absolute() → minimum_feedback_order() → score level, ascending

Graph: 538,263 edges (0 negative), largest strong component 11,972; DAG keeps 385,654 edges.

activation-order: Activating edges only, earliest in acyclic order

keep_sign(sign=1) → absolute() → minimum_feedback_order() → score level, ascending

Graph: 1,787,858 edges (0 negative), largest strong component 11,988; DAG keeps 1,299,978 edges.

repression-order: Repressing edges only, earliest in acyclic order

keep_sign(sign=-1) → absolute() → minimum_feedback_order() → score level, ascending

Graph: 537,315 edges (0 negative), largest strong component 10,626; DAG keeps 465,019 edges.

positive-out-degree: Most positive outgoing edges

score positive_out_degree, descending

Graph: 2,325,173 edges (537,315 negative), largest strong component 11,992.

negative-net-flow: Negative outgoing minus incoming weight

score negative_net_flow, descending

Graph: 2,325,173 edges (537,315 negative), largest strong component 11,992.

sign-aware: Rank sum: positive out-degree and negative net flow

score sign_aware, descending

Graph: 2,325,173 edges (537,315 negative), largest strong component 11,992.

reverse-pagerank: Highest reverse PageRank (raw graph)